EDC459: Week 7
SQL, Data Tidying & Assessment Preparation

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Sign In

Learning Objectives

What will I learn? How does this week fit into my course?

By the end of this week you will:

- Reinforce your understanding of good data and project management for reproducible data analysis by having a tested and working R project directory set up for your assessment.
- Gained experience connecting securely to and querying a real relational database hosted on an external server.
- Have a conceptual and practical understanding of essential data tidying skills, having applied selects, filters, mutates, pivots, and joins in both **R** and **SQL**.
- Have put these skills together in order to build a reproducible analytical pipeline that a) connects to an external database, b) queries and joins relational data, c) tidies data and reformats it from long format to wide format, d) visualises and analyses the data in line with the tasks in Part A of the assessment, e) produces a shareable PDF report using RMarkdown.

Today's session is designed to be more interactive and hands-on than our other sessions. We'll mostly be working together through a series of lecture notes; these are available on Blackboard.

Data tidying requires the use of multiple tools

- The idea of today is to introduce you to all the tools that will allow you to tidy any untidy dataset.
- You won't be able to use all of them perfectly right from the start.
- But if you invest time into learning to use them, you can become very proficient in data tidying.



Why spend time learning how to tidy data?

- Tidying data and preparing it for analysis or visualisation is often the most time consuming part of any quantitative research project.
- Tidying data is not often reproducable *unless it has been tidied programatically*.



Why SQL?

- We could just go through how to tidy everything in R — some of the tools we're talking about today we've already been using.
- But, Structured Query Language (SQL) is a skill that a large number of employers expect for quantitative research, data analyst, and data science roles.
- It is not that difficult to learn, but it is hard to find straightforward and realistic ways to practice it outside of learning 'on-the-job' unless you are also quite tech-minded and can set up your own local database or server.



Go back

University College London Hospitals NHS Foundation Trust

Senior Research Data Analyst

The closing date is 09 November 2026

Apply for this job

Job summary

Are you an experienced data analyst with knowledge of healthcare research? Do you have experience of the UK regulatory environment as it applies to Healthcare research?

Do you have the drive and commitment to overcome data quality issues and to improve working practice? If so, University College London Hospitals NHS Foundation Trust would love to hear from you.

Main duties of the job

The role will be responsible for all aspects of data collection and providing data reporting and analytics support to the directorate. The post-holder will play a key role in the collection, management and reporting of high-volume research data, serving a range of different purposes.

For questions about the job, contact:

Research Data and Information Manager

Stuart Braverman

s.braverman@nhs.net

NHS University College London Hospitals NHS Foundation Trust

Date posted

26 October 2026

Experience, Skills & Abilities

Essential

- Experience of relational databases
- In depth specialist knowledge of information analysis and the use of information across the research pathway acquired through degree or equivalent experience.
- Data management and reporting experience
- Developing the technical skills of others

Desirable

- An understanding of the mechanisms employed for financial control and economy

Skills and Abilities

Essential

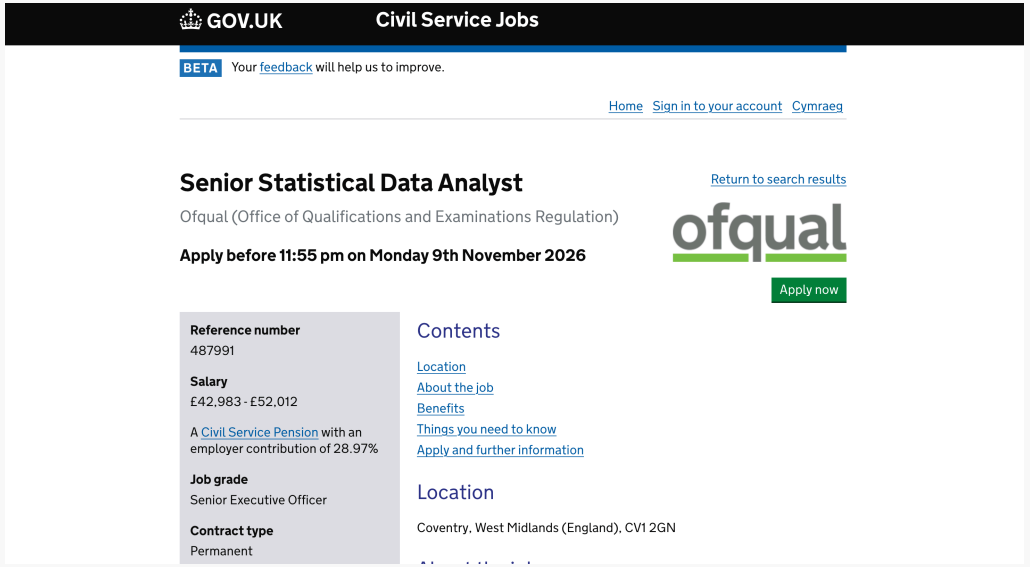
- Ability to convert data to management information
- Highly data rational with the ability to use quantitative and qualitative information to stimulate improvement

Values

Essential

- Demonstrate ability to meet Trust Values of safety, kindness, teamwork and improving

Communication



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Senior Statistical Data Analyst

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Ofqual (Office of Qualifications and Examinations Regulation)

ofqual

Apply before 11:55 pm on Monday 9th November 2026

Apply now

Reference number
487991

Salary
£42,983 - £52,012

A Civil Service Pension with an employer contribution of 28.97%

Job grade
Senior Executive Officer

Contract type
Permanent

Contents

Location
About the job
Benefits
Things you need to know
Apply and further information

Location

Coventry, West Midlands (England), CV1 2GN

Should a large number of applications be received, an initial sift will be undertaken using the work history and personal statement against the lead essential criteria, which is fluency in the R statistical programming language, applying this to analyst and interpret large data sets. Some experience of other similar languages also desirable (for example, Python, SQL, SAS).

Camden

Data Analyst (SEND)

★ bookmark

Camden Council | London Borough of Camden

On-Site | Expiring in less than 4 weeks

Data Analyst + | Full time | £52,205 per year

Data visualisation: You will use the most appropriate medium to visualise data to tell compelling stories that are relevant to business goals and can be acted upon. Your work will take advantage of a wide variety of data visualisation tools and methodologies, presenting complex information in a way that is engaging, useful and readily intelligible to a range of audiences such as frontline staff, managers, and senior leadership. You will present, communicate, and disseminate data appropriately and with influence in settings ranging from operational meetings to high profile strategic partnerships.

IT and mathematics: You will apply your knowledge and experience of IT and mathematical skills, including tools and techniques. You can adopt those most appropriate for the environment and always work in a manner that is sensitive to information security. You will use your experience of using a variety of tools such as MS Excel, Qlik, PowerBI, SQL, R, Python, QGIS.

Logical and creative thinking: You will respond effectively to problems in databases, esses, data products and services as they occur. You will initiate actions, monitor services, and identify trends to resolve problems. You will take responsibility for determining innovative solutions and assist with their implementation, and with preventative measures. Your analysis will involve creative interaction with data that takes innovative approaches to generating insight and answering business questions.

Lead Business Intelligence Analyst

Save this job

Royal Papworth Hospital NHS Foundation Trust

Cambridge Bio Medical Campus CB2 0AY, Huntingdon PE29 6FH

Salary: **£49,387 to £56,515 a year**

Date posted: **6 November 2026**

Closing date: **19 November 2026**

Contract type: **Fixed-Term**

Working pattern: **Full time**

Desirable

- Ability to write detailed analytic queries in SQL, investigate and resolve complex queries, issues and problems.
- Experience in acute RTT/Cancer/Diagnostics Reporting
- Knowledge of and ability to utilise statistical reporting solutions such as `R` and `SPSS`
- Experience of leading a team.
- Able to demonstrate an ability to move between the detail and the big picture;
- Knowledge and understanding of NHS data flows and data sets available within the NHS, including knowledge of the NHS Data Dictionary, currencies of healthcare commissioning, coding and information systems within the NHS;
- Experience of working with databases, especially Microsoft SQL Server;
- Experience of building reporting dashboards in Qlikview
- Coaching and mentoring others with analytical tasks

Senior Research Data Analyst

Save this job

University College London Hospitals NHS Foundation Trust

London NW1 2PG

Salary: **£47,951 to £56,863 a year**

Date posted: **26 October 2026**

Closing date: **9 November 2026**

Contract type: **Permanent**

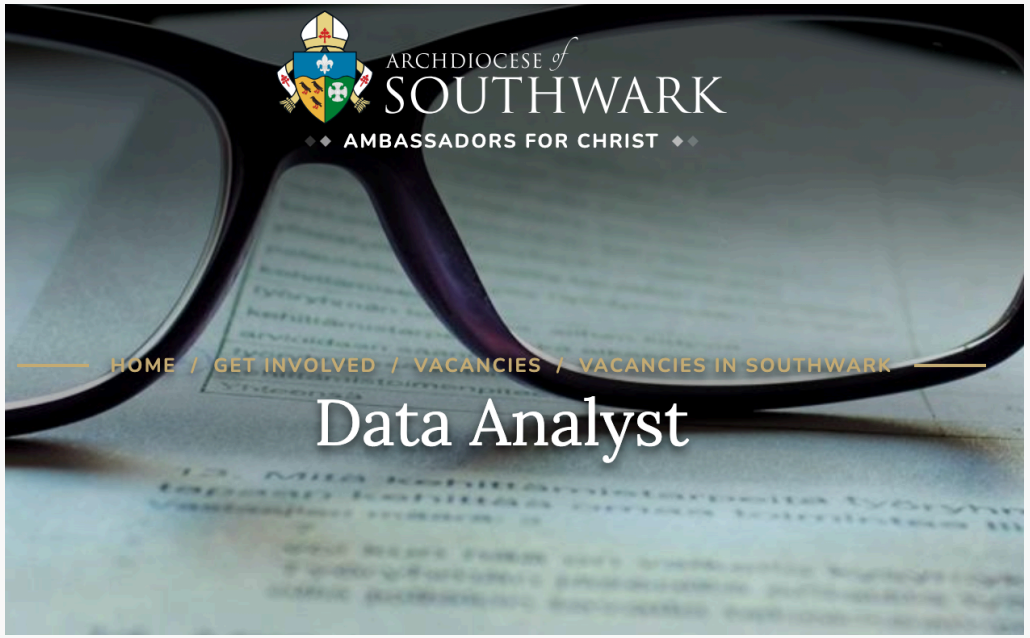
Working pattern: **Full time**

Person Specification

Knowledge and Qualifications

Essential

- Diploma in Information Technology or Data Analysis or equivalent experience
- Educated to degree level or equivalent experience.
- Ability to interrogate databases
- In depth knowledge of research workflows, especially those relating to Research Management and Governance
- Understanding of the need to maintain confidentiality and security in handling data according to the policies of the trusts and the Data Protection Act



Qualifications

Essential

- Degree, or equivalent qualification, in a numerate, data or information-related discipline — or equivalent demonstrable experience
- Evidence of continued professional development

Desirable

- Professional qualification or accreditation in data analysis, statistics or a related field
- Certification in data tools such as Power BI or SQL

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Digital Analyst (FTC)

Apply

Stratford (2 Redman Place)

Full time

Posted 3 Days Ago

R034666

£35,000 - £39,000 per annum plus benefits
Reports to: Digital Analytics Manager
Directorate: Marketing, Fundraising & Engagement
Contract: 12 month fixed-term contract
Hours: Full time 35 hours per week

Digital Analyst (FTC)

Apply

- Investigate data quality issues and ensure reporting remains accurate and consistent.
- Contribute to the continuous improvement of analytics, experimentation and reporting practices across the organisation.

What are we looking for?

- Experience using Google Analytics 4 (GA4) to analyse website performance and user behaviour.
- Experience designing and developing reporting for A/B and multivariance testing programmes.
Experience in working with data from platforms such as Optimizely, Google Ads or similar.
- Strong analytical skills, with experience working with large datasets and tools such as SQL, Python or R.
- Experience creating dashboards and visualisations using tools such as Power BI and/or Looker Studio.
- Ability to translate complex data into clear recommendations for non-technical audiences.
- Strong organisation skills with the ability to manage multiple priorities and deliver to deadlines.
- Excellent stakeholder engagement and communication skills.
- Experience using Google BigQuery would be desirable

Our organisation values are designed to guide all that we do.

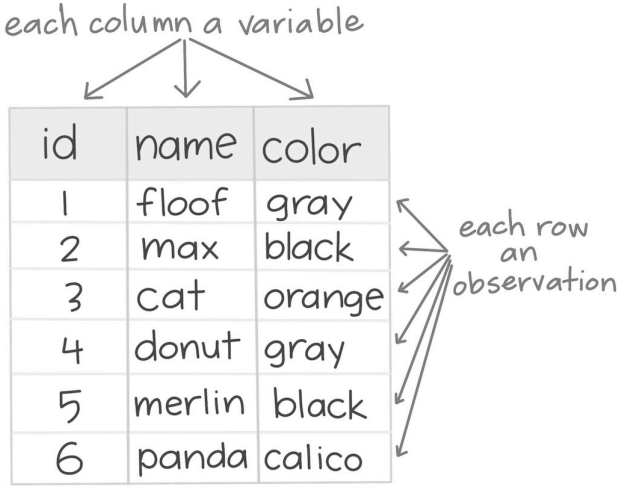
Introduction: What is tidy data?

"TIDY DATA is a standard way of mapping the meaning of a dataset to its structure."

–HADLEY WICKHAM

In tidy data:

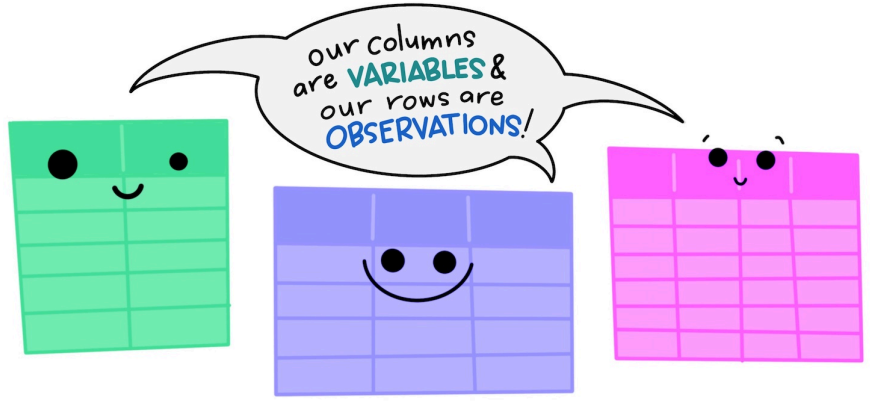
- each variable forms a column
- each observation forms a row
- each cell is a single measurement



Wickham, H. (2014). Tidy Data. Journal of Statistical Software *59* (10). DOI: 10.18637/jss.v059.i10

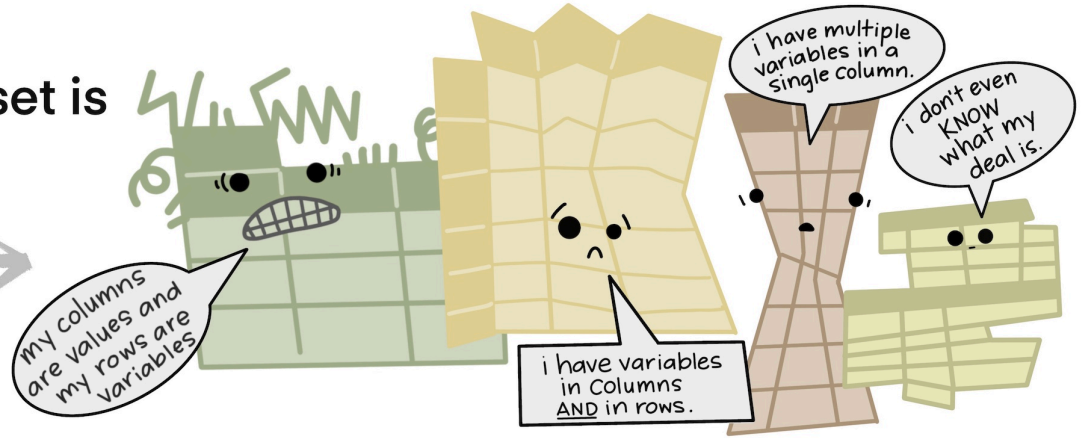
Illustrations from the Openscapes blog Tidy Data for reproducibility, efficiency, and collaboration by Julia Lowndes and Allison Horst

The standard structure of tidy data means that "tidy datasets are all alike..."



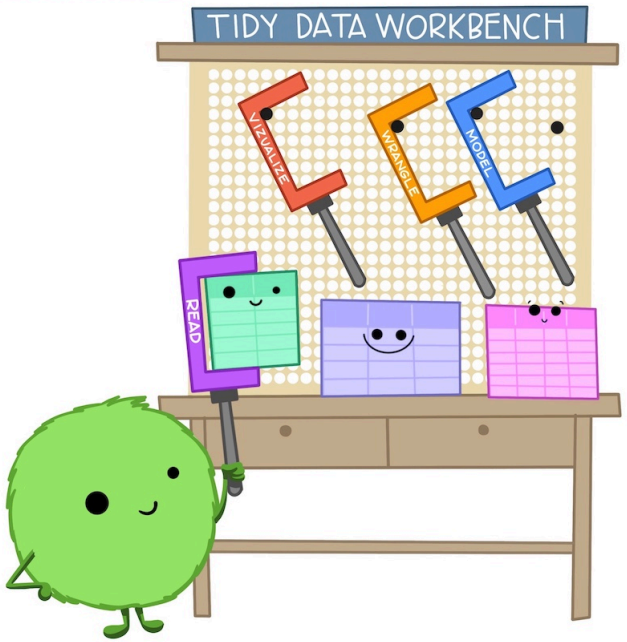
"...but every messy dataset is messy in its own way."

–HADLEY WICKHAM

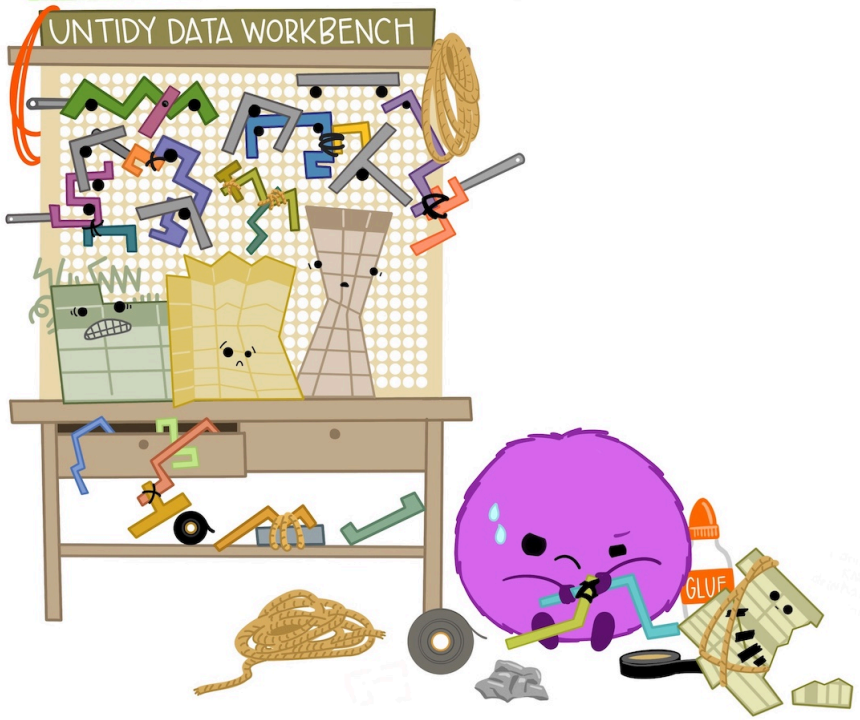


Illustrations from the Openscapes blog Tidy Data for reproducibility, efficiency, and collaboration by Julia Lowndes and Allison Horst

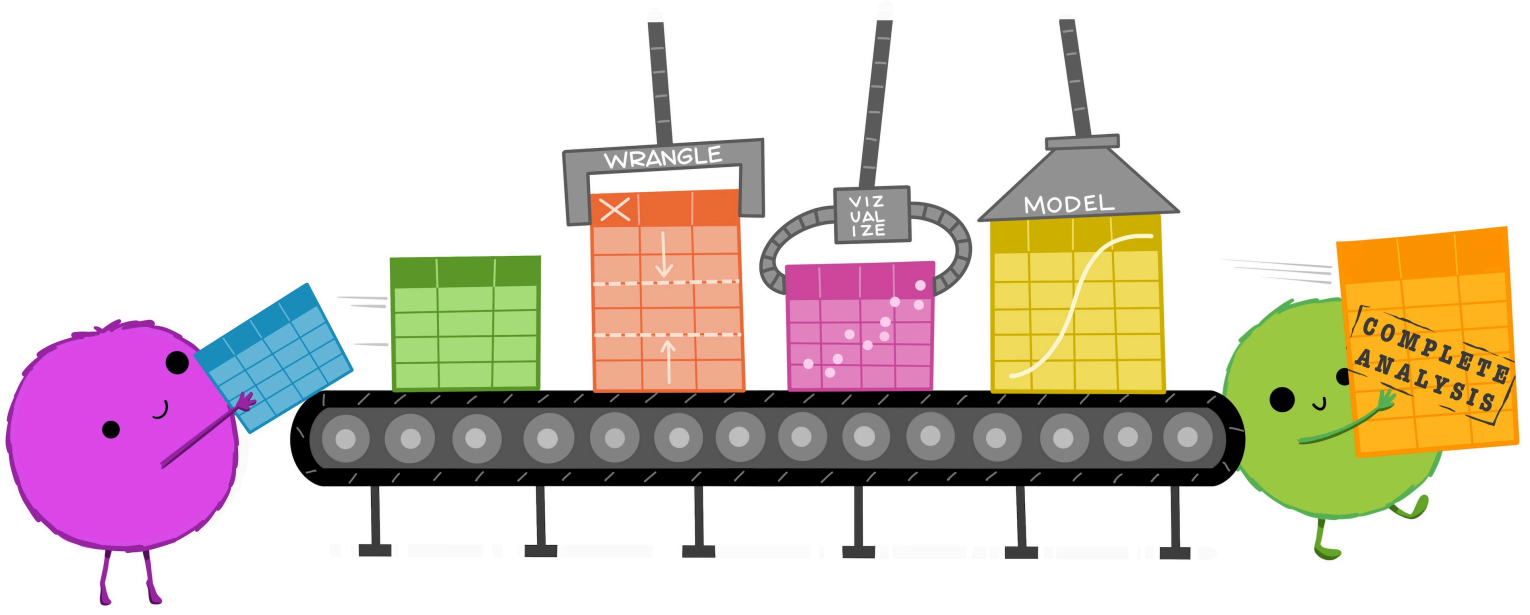
When working with tidy data, we can use the same tools in similar ways for different datasets...



...but working with untidy data often means reinventing the wheel with one-time approaches that are hard to iterate or reuse.



Illustrations from the Openscapes blog Tidy Data for reproducibility, efficiency, and collaboration by Julia Lowndes and Allison Horst



Illustrations from the Openscapes blog Tidy Data for reproducibility, efficiency, and collaboration by Julia Lowndes and Allison Horst

1. Reading data from different data sources and select/rename columns.

- Reading data from Stata, SPSS, and Excel files
- Filtering variables using select()
- Generalised select()

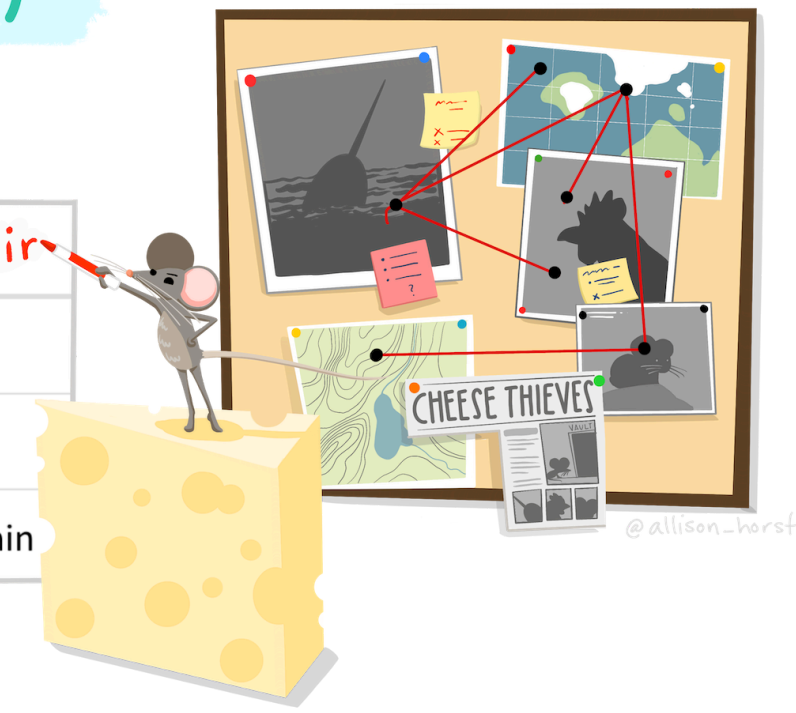
dplyr::rename()

RENAME COLUMNS*

```
df %>% rename(lair = site)
```

~~species~~ nemesis	status	~~site~~ lair
narwhal	unknown	ocean
chicken	active	coop
pika	active	mountain

*See **rename_with()** to rename using a function.



Artwork by @allison_horst



Artwork by @allison_horst

2. Creating new variables using mutate()

- Creating new variables that are transformations of existing variables
- Recoding categorical variables using case_when()
- Extracting numbers with parse_number()
- Performing repeated/generalised transformations



Artwork by @allison_horst

dplyr::case_when()

IF ELSE... (but you love it?)

```
df %>% mutate(danger = case_when(type == "kraken" ~ "extreme!",
                                 TRUE ~ "high"))
```

ADD COLUMN 'danger'

IF type is kraken THEN danger is extreme!

OTHERWISE, danger is high.



Artwork by @allison_horst



Artwork by @allison_horst

3. Aggregating data to higher levels with group_by()

- Creating new, aggregated datasets using group_by() and summarise()
- Adding group-level variables for multilevel models using group_by() and mutate()



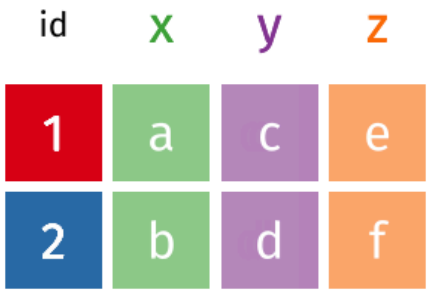

```
df |>
  group_by(cat1) |>
  summarize(
    avg = mean(x),
    total = sum(x),
    n = n()
  )
```

Animation by Andrew Heiss

4. Pivoting data between wide and long formats

- Converting wide datasets suitable for Latent Growth Structural Equation Modelling to long datasets suitable for multilevel modelling.
- ... and the reverse.

wide



Animation by Garrick Aden-Buie

wide

id	x	y	z
1	a	c	e
2	b	d	f

long

id	key	val
1	x	a
2	x	b
1	y	c
2	y	d
1	z	e
2	z	f

Animation by Garrick Aden-Buie

5. Working with strings and a little bit of regex

- How to remove certain characters or strings from character type variables (especially footnotes).
- Extracting subsets of characters from longer strings.
- Splitting variables into multiple columns based on a character within a string.



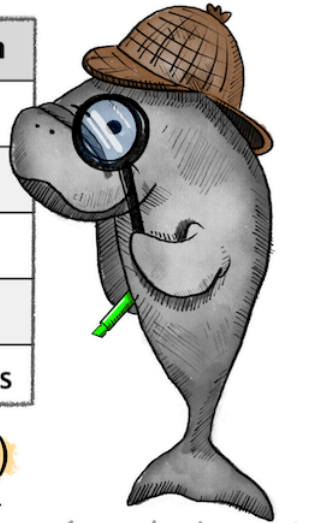
Artwork by @allison_horst

stringr::str_detect()

DETECT PRESENCE/ABSENCE* OF A PATTERN IN A STRING!

(df)

name	species	description
doug	dugong	awesome
olive	otter	super cool
greta	gorilla	superb
wilma	wolf	big, bad
barry	bonobo	superstitious



(a string detecting dugong)

Example:

```
str_detect(df$description, pattern = "super")
```

*add negate=TRUE to detect the absence of a pattern

Outcome:

```
FALSE  TRUE  TRUE  FALSE  TRUE
```

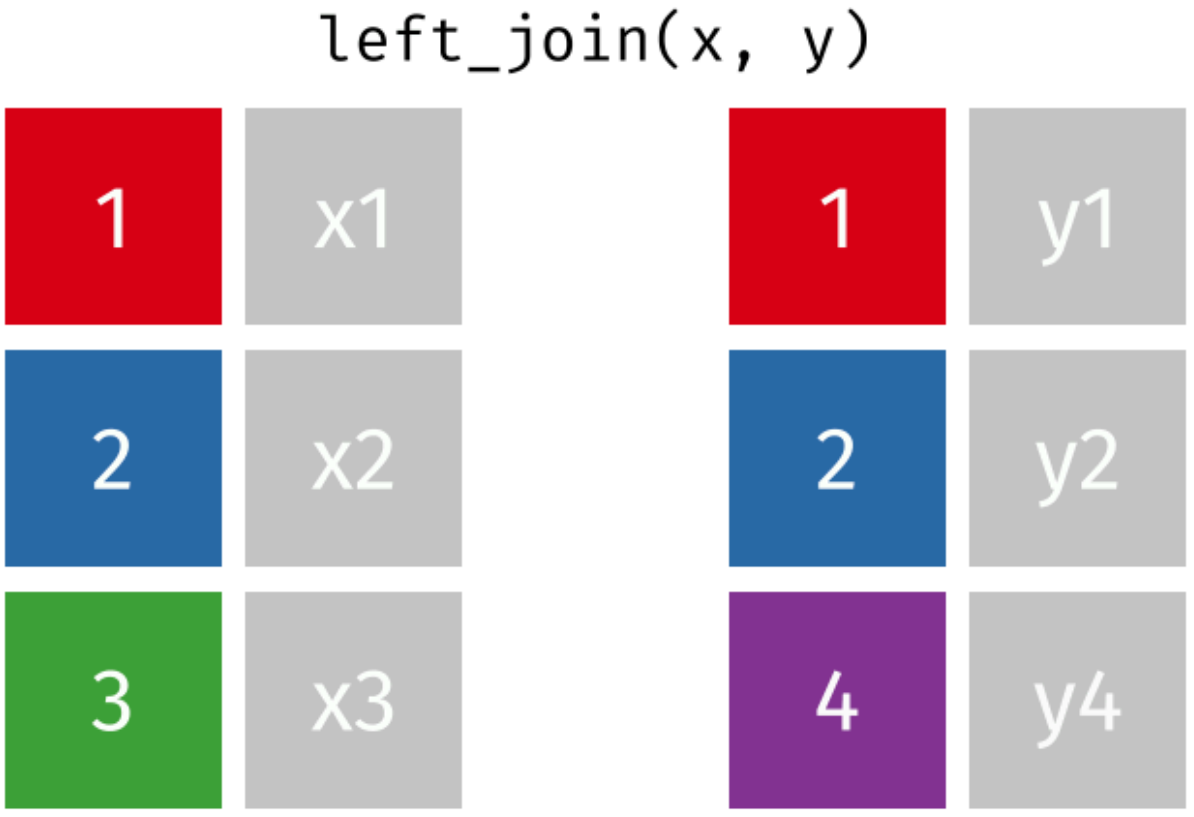
@allison_horst

Artwork by @allison_horst

6. Joining relational datasets

- Joining datasets based on a shared key
- Joining datasets together based on a combination of variables that form a key
- Joining higher level data to lower level data
- Checking for missing observations with anti_join()

Key: A value, usually a string, that uniquely identifies each observation across multiple related (relational) datasets



Animation by Garrick Aden-Buie

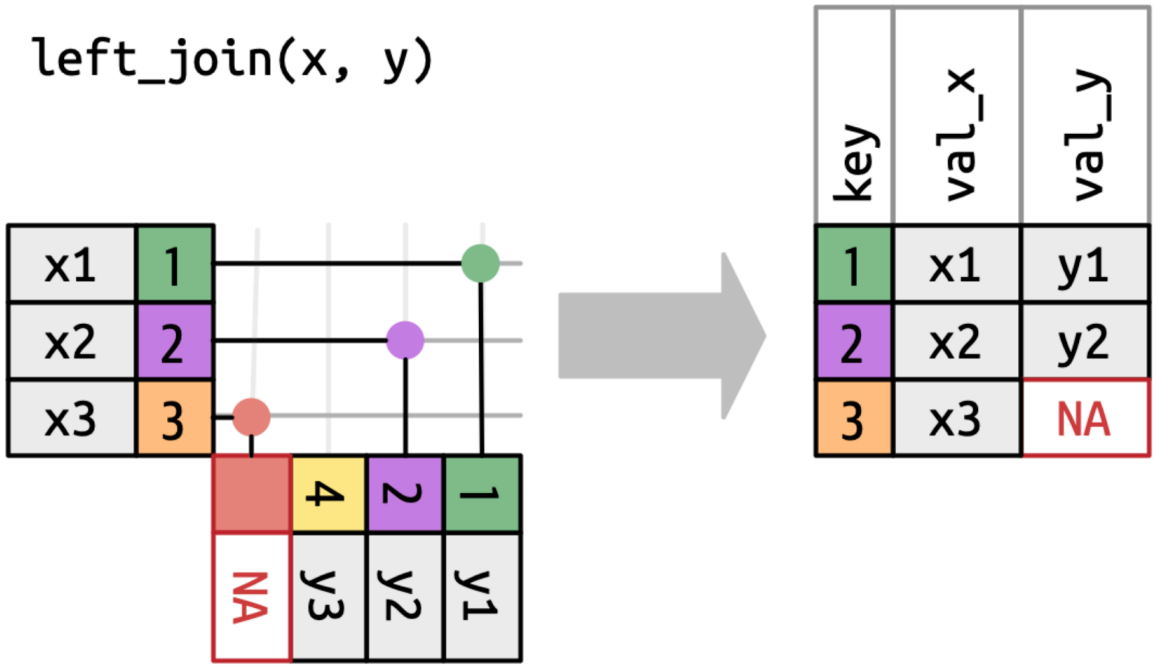


Diagram from R for Data Science 2e

7. Working with dates

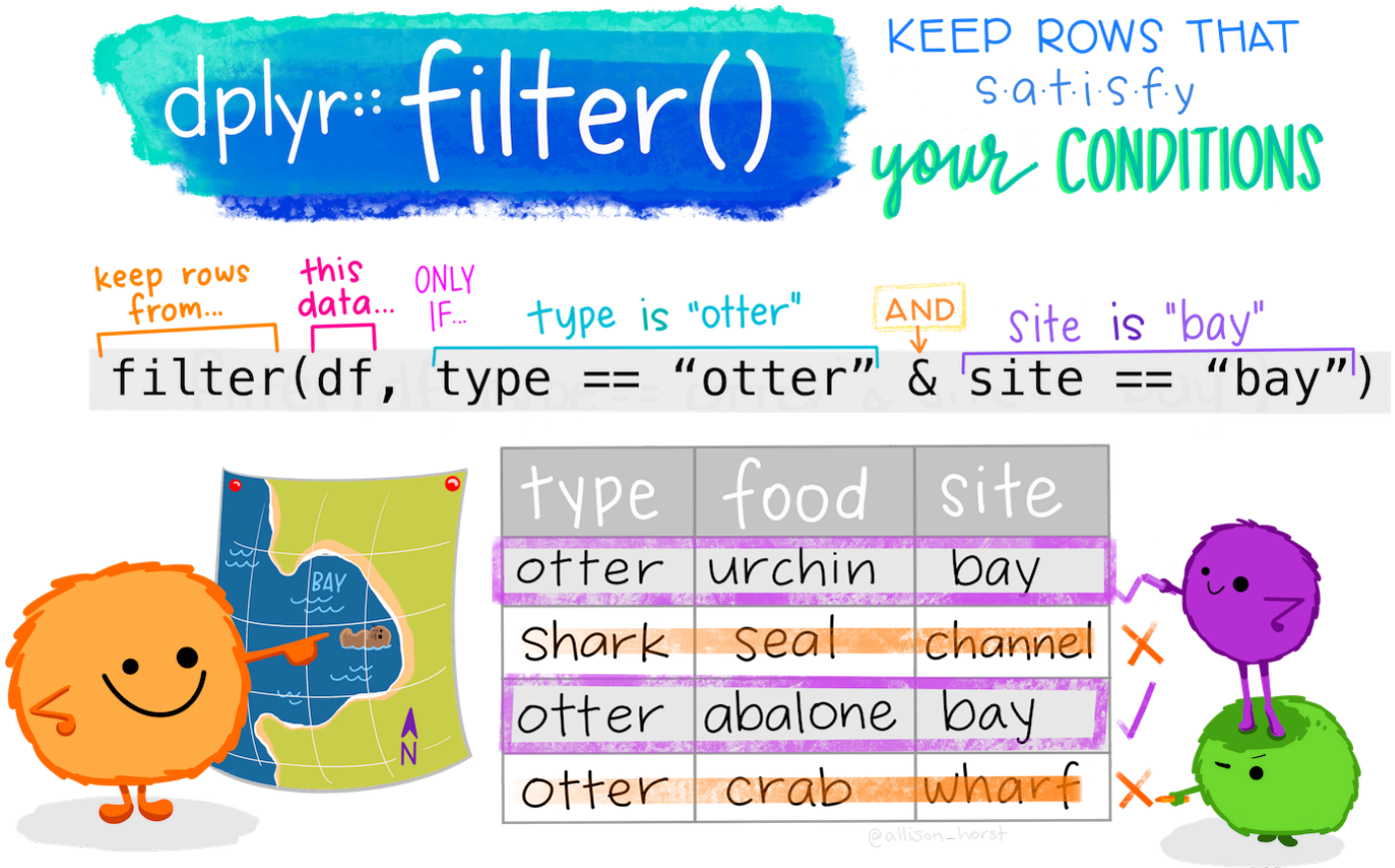
- How to fix how R interprets dates when they aren't in YYYY-MM-DD format.



Artwork by @allison_horst

8. Filtering rows of observations

- How to filter data based on values in character/factor type variables
- How to filter data based on numeric type variables
- How to filter data based on dates



Artwork by @allison_horst

That's a preview of some of the tools we'll be using today.

- Don't expect to be perfect at this just from a few attempts! I have the benefit of over ten years experience tidying datasets.
- If you keep practicing tidying untidy datasets, using these tools will eventually become effortless. Untidy data becomes a puzzle to solve. But when you're just starting out, the puzzles will be frustrating.



Today's practical session

I am going to be working through the process of setting up our R project for our assessment, testing our RMarkdown files knit and that you know how to complete the coversheet and MySkills evidence, and then showing you how to use data tidying and reading tools in R and SQL. We'll then do a "fake" Part A assessment together using the data we tidied.

